

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY (ICDFA)

Lab 6 - Firewall Traffic Control and Forensic Verification

Lab Title:	Firewall Traffic Control and Forensic Verification
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Student ID:	2025/FWSD/11339
Course Code:	SBT-DF203
Course Name:	Basic Networking Skills for Digital Forensics

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1. Required Tools and Environment

Item	Recommended Tool	Purpose
Server VM	Kali/Ubuntu with Apache	Protected host.
Client VM	Windows on host-only network	Generate HTTP requests.
Firewall tools	iptables, iptables-save	Apply and preserve rules.
Capture tools	tshark/Wireshark	Capture baseline and blocked traffic.
Client tools	curl, browser	Test access.
Optional extension	NetfilterQueue	Instructor-led packet interception demonstration.

2. Lab Folder Set up and Baseline Evidence

```
(kali@kali)-[~/Desktop]
└─$ mkdir -p ~/SBT-DF203-Lab6/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab6
pwd
/home/kali/SBT-DF203-Lab6
find . -maxdepth 1 -type d -print
./exported
./evidence
./working
./screenshots
./scripts
./reports
```

```
Unpacking apache2-data (2.4.68-2) over (2.4.68-1)...
Preparing to unpack ../apache2-utils_2.4.68-2_amd64.deb...
Unpacking apache2-utils (2.4.68-2) over (2.4.68-1)...
Setting up apache2-bin (2.4.68-2)...
Setting up apache2-data (2.4.68-2)...
Setting up apache2-utils (2.4.68-2)...
Setting up apache2 (2.4.68-2)...
apache-htcacheclean.service is a disabled or a static unit not running, not starting it.
Processing triggers for kali-menu (2026.3.2)...
Processing triggers for man-db (2.13.1-1)...
Scanning processes ...
Scanning linux images ...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.

(kali@kali)-[~/SBT-DF203-Lab6]
└─$ printf '<!DOCTYPE html>\n<html><body><h1>ICDFA Network Forensics Firewall Lab</h1><p>Server: YOUR SERVER
NAME</p></body></html>\n' | sudo tee /var/www/html/firewall_lab.html
<!DOCTYPE html>
<html><body><h1>ICDFA Network Forensics Firewall Lab</h1><p>Server: YOUR SERVER
NAME</p></body></html>
```

```

(kali㉿kali)-[~/SBT-DF203-Lab6]
└─$ sudo iptables-save | tee reports/iptables_before.rules
sudo iptables -L -n -v --line-numbers | tee reports/iptables_before.txt
sha256sum reports/iptables_before.rules | tee reports/iptables_before_sha256.txt
[sudo] password for kali:
# Generated by iptables-save v1.8.13 (nf_tables) on Fri Sep 18 17:47:49 2026
*filter
:INPUT ACCEPT [0:0]
:FORWARD DROP [0:0]
:OUTPUT ACCEPT [0:0]
:DOCKER - [0:0]
:DOCKER-BRIDGE - [0:0]
:DOCKER-CT - [0:0]
:DOCKER-FORWARD - [0:0]
:DOCKER-ISOLATION-STAGE-1 - [0:0]
:DOCKER-ISOLATION-STAGE-2 - [0:0]
:DOCKER-USER - [0:0]
-A FORWARD -j DOCKER-USER
-A FORWARD -j DOCKER-FORWARD
-A DOCKER ! -i docker0 -o docker0 -j DROP
-A DOCKER-BRIDGE -o docker0 -j DOCKER
-A DOCKER-CT -o docker0 -m conntrack --ctstate RELATED,ESTABLISHED -j ACCEPT
-A DOCKER-FORWARD -j DOCKER-CT
-A DOCKER-FORWARD -j DOCKER-ISOLATION-STAGE-1
-A DOCKER-FORWARD -j DOCKER-BRIDGE
-A DOCKER-FORWARD -i docker0 -j ACCEPT
-A DOCKER-ISOLATION-STAGE-1 -i docker0 ! -o docker0 -j DOCKER-ISOLATION-STAGE-2
-A DOCKER-ISOLATION-STAGE-2 -o docker0 -j DROP
COMMIT
# Completed on Fri Sep 18 17:47:49 2026
# Generated by iptables-save v1.8.13 (nf_tables) on Fri Sep 18 17:47:49 2026

```

num	pkts	bytes	target	prot	opt	in	out	source	destination
Chain DOCKER (1 references)									
num	pkts	bytes	target	prot	opt	in	out	source	destination
1	0	0	DROP	all	--	!docker0	docker0	0.0.0.0/0	0.0.0.0/0
Chain DOCKER-BRIDGE (1 references)									
num	pkts	bytes	target	prot	opt	in	out	source	destination
1	0	0	DOCKER	all	--	*	docker0	0.0.0.0/0	0.0.0.0/0
Chain DOCKER-CT (1 references)									
num	pkts	bytes	target	prot	opt	in	out	source	destination
1	0	0	ACCEPT	all	--	*	docker0	0.0.0.0/0	0.0.0.0/0
Chain DOCKER-FORWARD (1 references)									
num	pkts	bytes	target	prot	opt	in	out	source	destination
1	0	0	DOCKER-CT	all	--	*	*	0.0.0.0/0	0.0.0.0/0
2	0	0	DOCKER-ISOLATION-STAGE-1	all	--	*	*	0.0.0.0/0	0.0.0.0/0
3	0	0	DOCKER-BRIDGE	all	--	*	*	0.0.0.0/0	0.0.0.0/0
4	0	0	ACCEPT	all	--	docker0	*	0.0.0.0/0	0.0.0.0/0
Chain DOCKER-ISOLATION-STAGE-1 (1 references)									
num	pkts	bytes	target	prot	opt	in	out	source	destination
1	0	0	DOCKER-ISOLATION-STAGE-2	all	--	docker0	!docker0	0.0.0.0/0	0.0.0.0/0
Chain DOCKER-ISOLATION-STAGE-2 (1 references)									
num	pkts	bytes	target	prot	opt	in	out	source	destination
1	0	0	DROP	all	--	*	docker0	0.0.0.0/0	0.0.0.0/0
Chain DOCKER-USER (1 references)									
num	pkts	bytes	target	prot	opt	in	out	source	destination
d7be54408265f52ea348a762d77967aaaa5de12f635db3c1ec6379e1170c7609									reports/iptables_before.rules

Activate Windows
 Go to Settings to activate Windows

```

(kali㉿kali)-[~/SBT-DF203-Lab6]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:e6:98:7a brd ff:ff:ff:ff:ff:ff
    inet 192.168.229.128/24 brd 192.168.229.255 scope global dynamic noprefixroute eth0
        valid_lft 1662sec preferred_lft 1662sec
    inet6 fe80::c48e:ee4c:6374:462d/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 46:96:e3:6a:6c:50 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever

```

Server ip

```

(kali㉿kali)-[~/SBT-DF203-Lab6]
$ sudo systemctl start apache2
sudo ss -ltnp | grep ':80'
[sudo] password for kali:
LISTEN 0      511          *:*          users:((("apache2",pid=8625,fd=4),("apache2",pid=8624,fd=4),("apache2",pid=8623,fd=4),("apache2",pid=8622,fd=4),("apache2",pid=8621,fd=4),("apache2",pid=1005,fd=4))

```

Apache at work

```

C:\WINDOWS\system32\cmd.exe
Microsoft Windows [Version 10.0.19045.7548]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Akanbi Initiatives>ping 192.168.229.128

Pinging 192.168.229.128 with 32 bytes of data:
Reply from 192.168.229.128: bytes=32 time<1ms TTL=64
Reply from 192.168.229.128: bytes=32 time<1ms TTL=64
Reply from 192.168.229.128: bytes=32 time<1ms TTL=64
Reply from 192.168.229.128: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.229.128:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

Server and client connetion

```
C:\WINDOWS\system32\cmd.exe

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter VMware Network Adapter VMnet1:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::4e58:1914:204b:e582%17
    IPv4 Address. . . . . : 192.168.239.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::1cc6:7529:cbe6:3970%16
    IPv4 Address. . . . . : 192.168.229.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::5cd0:1ba1:1ce3:9d94%8
    IPv4 Address. . . . . : 10.94.78.175
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.94.78.9
```

Client IP Address

3. Chain-of-Custody Worksheet

Field	Your Entry
Case/lab identifier	SBT-DF203-Lab6
Trainee name	Yusuf Oluwaseun Dada
Date and time started	16/07/2026 4:30pm
Evidence file name(s)	
Source or generation method	Self-generated (server VM + client VM on host-only network)
Original SHA-256	d7be54408265f52ea348a762d77967aaea5de12f635db3c1ec6379e1170c7609
Working-copy SHA-256	d7be54408265f52ea348a762d77967aaea5de12f635db3c1ec6379e1170c7609

4. Part A - Record the Lab Network and Baseline Access

```
(kali㉿kali) [~/SBT-DF203-Lab6]
ip -br address | tee reports/server_interfaces.txt
ip route | tee reports/server_routes.txt
sudo ss -lntp | grep ':80' | tee reports/apache_listener.txt
to UNKNOWN 127.0.0.1/8 ::1/128
eth0 UP 192.168.229.128/24 fe80::c48e:ee4c:6374:462d/64
docker0 DOWN 172.17.0.1/16
default via 192.168.229.2 dev eth0 proto dhcp src 192.168.229.128 metric 100
172.17.0.1/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown
192.168.229.0/24 dev eth0 proto kernel scope link src 192.168.229.128 metric 100
LISTEN 0 511 *:* *:* users:((("apache2",pid=8625,fd=4),("apache2",pid=8624,fd=4),("apache2",pid=8623,fd=4),("apache2",pid=8622,fd=4),("apache2",pid=8621,fd=4),("apache2",pid=1005,fd=4))
```

On the server

```
ip -br address | tee reports/server_interfaces.txt
```

```
ip route | tee reports/server_routes.txt
```

```
sudo ss -lntp | grep ':80' | tee reports/apache_listener.txt
```

```
(kali㉿kali)-[~/SBT-DF203-Lab6]
$ curl -v --connect-timeout 5 http://192.168.229.128/firewall_lab.html 2>&1 | tee baseline_curl.txt
* Trying 192.168.229.128:80 ...
* Established connection to 192.168.229.128 (192.168.229.128 port 80) from 192.168.229.128 port 60634
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total      Spent    Left     Speed
0          0  0       0  0      0         0      0      0      0      0      0      0      0
0* using HTTP/1.x
> GET /firewall_lab.html HTTP/1.1
> Host: 192.168.229.128
> User-Agent: curl/8.21.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Fri, 18 Sep 2026 22:02:17 GMT
< Server: Apache/2.4.68 (Debian)
< Last-Modified: Fri, 18 Sep 2026 22:02:05 GMT
< ETag: "6c-65bc90d0bbcc45"
< Accept-Ranges: bytes
< Content-Length: 108
< Vary: Accept-Encoding
< Content-Type: text/html
<
{ [108 bytes data]
100    108 100    108    0      0     4556    0      0
* Connection #0 to host 192.168.229.128:80 left intact
<!DOCTYPE html>
<html><body><h1>ICDFA Network Forensics Firewall Lab</h1><p>Server: Kali </p></body></html>
```

```
# On the client, replace SERVER IP
```

```
curl -v --connect-timeout 5 http://192.168.229.128/firewall_lab.html 2>&1 | tee
baseline curl.txt
```

Item	Your Value
Server IP	192.168.229.128
Allowed client IP	192.168.229.1
Blocked client IP (instructor-assigned)	192.168.229.1
Server interface name	eth0
Default gateway	192.168.229.2

5. Part B - Capture the Allowed HTTP Baseline

```
(kali㉿kali)-[~/SBT-DF203-Lab6]
└─$ sudo tshark -i eth0 -f "host 192.168.229.1 and tcp port 80" -a duration:35 -w evidence/http_blocked.pcapng &
sleep 3
# NOW run curl from the client (the timeout one)
wait
sha256sum evidence/http_blocked.pcapng | tee reports/http_blocked_sha256.txt
[1] 54995
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'

[1] + done          sudo tshark -i eth0 -f "host 192.168.229.1 and tcp
port 80" -a duration:35 -w
7d5eb227f2e36a8f9ac3dc15d1761a48f23d5984148814e84e2e42187e9ffd5a evi
dence/http_blocked.pcapng
```

```
sudo tshark -i eth0
-f "192.168.229.128 and tcp port 80" -a duration:30 -w
evidence/http_allowed.pcapng &
sleep 3
```

```
(kali㉿kali)-[~/SBT-DF203-Lab6]
└─$ sudo iptables -I INPUT 1 -s "192.168.229.1
" -p tcp --dport 80 -j DROP
sudo iptables -L -n -v --line-numbers | tee reports/iptables_after_a
dd.txt
sudo iptables -C INPUT -s "192.168.229.1
" -p tcp --dport 80 -j DROP
printf 'Rule verified present.\n' | tee reports/rule_verification.tx
t
iptables v1.8.13 (nf_tables): host/network `192.168.229.1
' not found
Try `iptables -h' or 'iptables --help' for more information.
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target    prot opt in     out     source
      destination

Chain FORWARD (policy DROP 0 packets, 0 bytes)
num  pkts bytes target    prot opt in     out     source
      destination
1      0      0 DOCKER-USER  all  --  *      *       0.0.0.0/0
      0.0.0.0/0
2      0      0 DOCKER-FORWARD  all  --  *      *       0.0.0.0/0
      0.0.0.0/0

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target    prot opt in     out     source
      destination

Chain DOCKER (1 references)
num  pkts bytes target    prot opt in     out     source
      destination
```

6. Part C - Apply and Verify the Blocking Rule

```
Chain DOCKER-FORWARD (1 references)
num  pkts bytes target     prot opt in     out     source
destination
1      0      0 DOCKER-CT  all  --  *      *      0.0.0.0/0
0.0.0.0/0
2      0      0 DOCKER-ISOLATION-STAGE-1 all  --  *      *      0
.0.0.0/0
0.0.0.0/0
3      0      0 DOCKER-BRIDGE all  --  *      *      0.0.0.0/0
0.0.0.0/0
4      0      0 ACCEPT    all  --  docker0 *      0.0.0.0/0
0.0.0.0/0

Chain DOCKER-ISOLATION-STAGE-1 (1 references)
num  pkts bytes target     prot opt in     out     source
destination
1      0      0 DOCKER-ISOLATION-STAGE-2 all  --  docker0 !docker0
0.0.0.0/0
0.0.0.0/0

Chain DOCKER-ISOLATION-STAGE-2 (1 references)
num  pkts bytes target     prot opt in     out     source
destination
1      0      0 DROP      all  --  *      docker0 0.0.0.0/0
0.0.0.0/0

Chain DOCKER-USER (1 references)
num  pkts bytes target     prot opt in     out     source
destination
iptables v1.8.13 (nf_tables): host/network `192.168.229.1
' not found
Try `iptables -h' or 'iptables --help' for more information.
Rule verified present.
```

```
sudo iptables -I INPUT 1 -s "192.168.229.1"
```

```
" -p tcp --dport 80 -j DROP
```

```
sudo iptables -L -n -v --line-numbers | tee reports/iptables_after_add.txt
```

```
sudo iptables -C INPUT -s "192.168.229.1"
```

```
" -p tcp --dport 80 -j DROP
```

```
printf 'Rule verified present.\n' | tee reports/rule_verification.txt
```

7. Part D - Capture Blocked Traffic and Rule Counters

```
(kali@kali)-[~/SBT-DF203-Lab6]
$ sudo iptables -I INPUT 1 -s "192.168.229.1" -p tcp --dport 80 -j DROP

sudo iptables -L INPUT -n -v --line-numbers
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target     prot opt in     out     source
destination
1      0      0 DROP      tcp  --  *      *      192.168.229.1
0.0.0.0/0
tcp dpt:80
```

```
(kali㉿kali)-[~/SBT-DF203-Lab6]
$ sudo tshark -i eth0 -f "host 192.168.229.1 and tcp port 80" -a duration:35 -w /http_blocked.pcapng &
sleep 3
[1] 34543
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
```

On server

```
sudo tshark -i eth0 -f "host 192.168.229.1 and tcp port 80" -a duration:35 -w
evidence/http_blocked.pcapng &
sleep 3
```

```
C:\Users\Akanbi Initiatives>curl -v --connect-timeout 10 http://192.168.229.128/firewall_lab.html
* Trying 192.168.229.128:80...
* Connected to 192.168.229.128 (192.168.229.128) port 80
* using HTTP/1.x
> GET /firewall_lab.html HTTP/1.1
> Host: 192.168.229.128
> User-Agent: curl/8.13.0
> Accept: */*
>
< HTTP/1.1 200 OK
< Date: Sat, 19 Sep 2026 21:03:47 GMT
< Server: Apache/2.4.68 (Debian)
< Last-Modified: Fri, 18 Sep 2026 22:02:05 GMT
< ETag: "6c-65bc90d0bbc45"
< Accept-Ranges: bytes
< Content-Length: 108
< Vary: Accept-Encoding
< Content-Type: text/html
<
<!DOCTYPE html>
<html><body><h1>ICDFA Network Forensics Firewall Lab</h1><p>Server: Kali </p></body></html>
* Connection #0 to host 192.168.229.128 left intact

C:\Users\Akanbi Initiatives>curl -v --connect-timeout 10 http://192.168.229.128/firewall_lab.html
* Trying 192.168.229.128:80...
* Connection timed out after 10005 milliseconds
* closing connection #0
curl: (28) Connection timed out after 10005 milliseconds
```

From blocked client during capture:

```
curl -v --connect-timeout 10 http://192.168.229.128/firewall_lab.html
wait
```

```
(kali㉿kali)-[~/SBT-DF203-Lab6]
$ sudo iptables -L INPUT -n -v --line-numbers | tee reports/iptables_
s_after_test.txt
sudo sha256sum evidence/http_blocked.pcapng | tee reports/http_blocke
d_sha256.txt
[sudo] password for kali:
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target      prot opt in      out      source
destination
35cf8ef2e5156b6fa43659de7aa28d90e72f658701e6b02aa3fe9f60f2cb3455  evi
dence/http_blocked.pcapng
```

```
sudo iptables -L INPUT -n -v --line-numbers | tee reports/iptables_after_test.txt
Sudo sha256sum evidence/http_blocked.pcapng | tee reports/http_blocked_sha256.txt
```

8. Part E - Compare Allowed and Blocked Captures

```
(kali㉿kali)-[~/SBT-DF203-Lab6]
$ for PCAP in evidence/http_allowed.pcapng evidence/http_blocked.pcapng; do
  echo "==== $PCAP ===="
  tshark -r "$PCAP" -Y 'tcp.flags.syn==1 || http.request || http.response || tcp.analysis.retransmission' \
    -T fields -e frame.number -e frame.time_relative -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport \
    -e tcp.flags -e tcp.analysis.retransmission -e http.request.uri -e http.response.code
done | tee reports/allowed_vs_blocked.tsv
==== evidence/http_allowed.pcapng ====
==== evidence/http_blocked.pcapng =====
```

```
for PCAP in evidence/http_allowed.pcapng evidence/http_blocked.pcapng; do
  echo "==== $PCAP ====="
  tshark -r "$PCAP" -Y 'tcp.flags.syn==1 || http.request || http.response || tcp.analysis.retransmission' \
    -T fields -e frame.number -e frame.time_relative -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport \
    -e tcp.flags -e tcp.analysis.retransmission -e http.request.uri -e http.response.code
done | tee reports/allowed_vs_blocked.tsv
```

9. Part F - Remove the Rule and Restore Access

```
(kali㉿kali)-[~/SBT-DF203-Lab6]
$ sudo iptables -D INPUT -s "192.168.229.1" -p tcp --dport 80 -j DROP
sudo iptables -C INPUT -s "192.168.229.1" -p tcp --dport 80 -j DROP | echo 'Rule successfully removed'
sudo iptables -L INPUT -n -v --line-numbers | tee reports/iptables_restored.txt
iptables: Bad rule (does a matching rule exist in that chain?).
Rule successfully removed
Chain INPUT (policy ACCEPT 6 packets, 1061 bytes)
num  pkts bytes target    prot opt in     out     source
destination
```

BLOCKED_CLIENT_IP=192.168.229.1

```
sudo iptables -D INPUT -s "192.168.229.1" -p tcp --dport 80 -j DROP
```

```
sudo iptables -C INPUT -s "192.168.229.1" -p tcp --dport 80 -j DROP || echo 'Rule successfully removed'
```

```
sudo iptables -L INPUT -n -v --line-numbers | tee reports/iptables_restored.txt
```

```

C:\Users\Akanbi Initiatives>curl -v --connect-timeout 5 http://192.168.229.128/firewall_lab.htm
* Trying 192.168.229.128:80...
* Connected to 192.168.229.128 (192.168.229.128) port 80
* using HTTP/1.x
> GET /firewall_lab.htm HTTP/1.1
> Host: 192.168.229.128
> User-Agent: curl/8.13.0
> Accept: */*
>
< HTTP/1.1 404 Not Found
< Date: Sat, 19 Sep 2026 21:54:18 GMT
< Server: Apache/2.4.68 (Debian)
< Content-Length: 317
< Content-Type: text/html; charset=iso-8859-1
<
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html><head>
<title>404 Not Found</title>
</head><body>
<h1>Not Found</h1>
<p>The requested URL was not found on this server.</p>
<hr>
<address>Apache/2.4.68 (Debian) Server at 192.168.229.128 Port 80</address>
</body></html>
* Connection #0 to host 192.168.229.128 left intact

```

Confirm access from the previously blocked client

curl -v --connect-timeout 5 http://192.168.229.128/firewall_lab.html

10. Forensic Interpretation Questions

Question	Your Answer
1. DROP → retransmissions/timeout, why?	DROP silently discards the packet, which the client gets no reply at all. TCP can't tell the difference between "packet lost" and "deliberately blocked," so it assumes loss and retransmits the SYN a few times using normal backoff, then finally gives up and times out.
2. REJECT vs DROP difference	REJECT sends an explicit response back (usually a TCP RST or an ICMP "port unreachable"), so the client fails immediately with a clear error like "connection refused." DROP sends nothing back, so the client has to wait through several retries and a full timeout before giving up, this difference is visible in the capture as "RST present" (REJECT) vs "no response at all, just repeated SYNs" (DROP).
3. Why firewall counters matter	The rule listing only proves a rule exists, it doesn't prove it ever did anything. The packet/byte counter increasing from 0 to a real number is independent proof that actual traffic matched and triggered the rule, not just that the rule is sitting there unused.
4. Server-down vs firewall-blocked evidence	If the server itself is down (like not listening), the client usually gets an immediate "connection

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	refused" (an RST comes back right away, because the OS answers even though nothing is listening). If a firewall silently drops the traffic, the client gets no response at all and has to wait out a timeout. So the key evidence is: fast RST = service problem; long silence + timeout = firewall DROP.
5. Risk of deleting by line number	Line numbers shift any time a rule is added, removed, or reordered. If "rule #1" is deleted by number after the list has changed, it might delete the wrong rule entirely, possibly opening a hole you didn't mean to, or leaving the real DROP rule in place. Deleting by full match criteria (source IP, protocol, port, target) instead guarantees to remove the exact rule, regardless of its position in the list.